

Scientific messaging by academics and population mobility during the COVID-19 pandemic: the moderating role of cultural factors

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Abstract

Academics became more visible communicators during the COVID-19 pandemic and many engaged directly with public audiences through social media. We examine whether the daily share of academics' Twitter activity devoted to COVID-related scientific messaging is associated with subsequent changes in population mobility across 42 countries during the early phase of the pandemic, and whether the strength of this association varies across cultural contexts. Using a daily panel of 28,985,603 tweets from 244,132 academics between February 15 and December 31, 2020, we estimate panel linear regressions with country and week fixed effects, adjusting for policy stringency, the epidemiological situation, and overall academic Twitter activity. We test cultural moderation through interaction terms with each of the six Hofstede cultural dimensions.

We find that, all else equal, days when a higher share of academics' tweets contain scientific messaging are associated with reductions in out-of-home mobility (specifically retail and recreation, parks, transit stations, and workplaces) and increases in residential mobility. The association is most consistent for workplace mobility and emerges at varying lag lengths across other categories. The magnitude and significance of these associations vary across cultural contexts, with stronger negative associations observed in countries scoring higher on power distance and uncertainty avoidance, weaker associations in countries with higher long-term orientation, and more selective moderation by individualism and motivation towards achievement and success.

These findings suggest that scientific communication by academics during the pandemic appears to be embedded in a broader information environment associated with behavioral change, and that its relationship with population mobility varied across cultural settings.

Keywords: *COVID-19, science communication, human mobility, social media, culture, panel regression*

1 Introduction

The COVID-19 pandemic has substantially affected the interaction between science, policy, and society. Governments enacted containment measures, from lockdowns to travel restrictions, that demanded public compliance and required policymakers to work with scientific experts more closely than at any other time in recent history (Ball, 2021). The pandemic blurred the boundaries between science and public policy and elevated many experts to highly visible roles (Leidecker-Sandmann et al., 2022; Post et al., 2021). In this setting, scientific research unfolded under public and political scrutiny, and policy decisions relied on rapid updating of evidence. Effective communication thus became a matter of urgency for both institutional decision-making and population health.

As daily routines changed and in-person interactions were restricted, digital platforms played a larger role in how people accessed information and communicated with one another (Manuti et al., 2022). One development was the increasing use of social media by academics as a public space to compensate for limited physical day-to-day interactions (Nguyen et al., 2020). On Twitter (now X), for example, experts voiced personal views and offered fact-based recommendations, while engaging with audiences beyond their academic peers (Della Giusta et al., 2021; Rauchfleisch et al., 2021b). Many built substantial followings over the course of the pandemic (Tardy, 2023; Utz et al., 2022), reflecting the public’s appetite for credible information, as well as the platform’s ability to provide academics with a voice. Whether this kind of direct communication is associated with public behavior, however, remains mostly untested. One source of evidence comes from traditional media. Chan et al. (2023) showed that media framing of science in the UK was associated with subsequent changes in human mobility during the pandemic. However, traditional media introduces layers of mediation between academics and the public (Peters, 2013), and only select experts are typically featured (Rauchfleisch et al., 2021a; Schäfer, 2011). Whether similar patterns hold on social media, where academics communicate without intermediaries, and whether they extend across national contexts, is still an open question.

These questions are relevant to population health because mobility was one of the main behavioral pathways through which societies responded to COVID-19. During the pre-vaccine phase of the pandemic, reducing movement in public spaces was central to limiting interpersonal contact and slowing transmission (Nouvellet et al., 2021; Yan et al., 2021). Yet these changes could not be explained by formal policy measures alone, as people also changed their mobility behavior voluntarily in response to perceived epidemic risk (Yan et al., 2021). Prior work shows that trust in science and public health authorities was associated with better COVID-19 knowledge and stronger support for containment measures (Battiston et al., 2021), while social media became one channel through which health experts and authorities communicated such information to the public (Reveilhaç, 2022). If academics’ scientific messaging is associated with subsequent mobility changes, then it may capture part of the expert information environment that accompanied population-level behavioral adjustment during a public health crisis.

Whether these associations are uniform across societies is a separate question. Societies develop distinct cultural norms that shape what behaviors are considered appropriate, and these norms can lead to different responses to the same external conditions (Hofstede, 2011). In a pandemic, it is plausible that populations differ in how readily they act

on expert guidance and how much authority they accord to institutional voices in cases of uncertainty. Cross-cultural work on health attitudes and behavior during COVID-19 has explored these, with societies varying in their acceptance or support of pharmaceutical measures such as vaccines (Lu, 2023; Lamot and Kirbiš, 2024; Leonhardt and Pezzuti, 2022) and non-pharmaceutical measures such as quarantine (Ma et al., 2022) and mask-wearing (Lu et al., 2021). If the link between scientific messaging and mobility is conditioned by these underlying cultural dispositions, then a cross-country design provides a useful setting in which to examine such heterogeneity.

In this paper, we address two research questions. First, we explore whether the relative prevalence of COVID-related scientific messaging in academics’ daily Twitter activity is associated with subsequent changes in population mobility across countries during the early phase of the pandemic. We examine this association across six types of mobility (retail and recreation, grocery and pharmacy, parks, transit stations, workplaces, and residential movement), which allows us to distinguish between categories where individuals have more behavioral discretion, such as leisure or shopping, and those shaped by structural constraints, such as workplace attendance. We also consider multiple temporal lags (1, 7, and 14 days) to capture whether any association unfolds immediately or over a longer period. Second, we ask whether the strength of this association depends on cultural context. To examine this, we test the moderating role of each of the six cultural dimensions proposed by Hofstede (2011): power distance (PDI), individualism (IDV), motivation towards achievement and success (MAS), uncertainty avoidance (UAI), long-term orientation (LTO), and indulgence (IVR).

We assemble a cross-country panel of daily observations from February 15 to December 31, 2020 across 42 countries. This ten-month window captures the pre-vaccine phase of the pandemic, during which scientific communication was mainly concerned with non-pharmaceutical interventions such as masks, distancing, ventilation, and stay-at-home guidance (Hale et al., 2021; Pollett and Rivers, 2020).

We use two main data sources. For the dependent variable, we use Google’s COVID-19 Community Mobility Reports (Aktay et al., 2020), which provided daily mobility changes relative to pre-pandemic baselines across multiple trip categories. For the main independent variable, we collected aggregated Twitter data covering 28,985,603 tweets from 244,132 academics identified through a systematic sampling approach. To examine temporal associations, we estimate panel linear regression models with country and week fixed effects, adjusting for the epidemiological situation, formal policy environment, and overall levels of academic Twitter activity.

Our findings aim to inform a more specific understanding of how scientific communication relates to public behavior during health crises, and to offer evidence on the increasingly influential role of academics as communicators in the digital age.

We organize the rest of this paper as follows. Section 2 discusses the relevant literature. Section 3 outlines the research questions. Section 4 describes the data and methods. Section 5 presents the results. We close with the conclusions and limitations in Section 6 and Section 7, respectively.

2 Literature review

2.1 Academic engagement and science communication during the pandemic

Trust in scientific expertise becomes especially important in public health crises, where individuals often lack the knowledge or direct means to evaluate emerging risks on their own (Siegrist and Cvetkovich, 2000). In the early stages of an outbreak, pharmaceutical tools such as vaccines are rarely available, and containment depends mainly on behavioral change (Bavel et al., 2020).

The COVID-19 pandemic therefore created an unusually intense setting for public scientific communication. As the virus spread rapidly and governments faced urgent decisions under conditions of uncertainty, academics were drawn into fast-moving discussions about risk, evidence, and appropriate responses. Researchers took on roles beyond the traditional dissemination of findings, as they explained the reasoning behind recommendations, corrected misreadings of emerging evidence, and countered misinformation as it spread (Pollett and Rivers, 2020), particularly as conspiracy theories and health-related falsehoods were widely circulated on social media (Bavel et al., 2020).

Much of this communication unfolded on Twitter. The platform’s real-time structure suited the pace of the pandemic, as it allowed credible voices to reach large audiences and circulate evidence almost as quickly as it emerged (Lorenzoni et al., 2025; Tardy, 2023; Utz et al., 2022). Researchers used it to share preliminary findings and organize informal collaborations across institutional lines (Pollett and Rivers, 2020). At the same time, the directness that made the platform useful also exposed academics to new pressures. For example, findings could be misread or reshared without proper context (Pollett and Rivers, 2020), and scientific messages were easily absorbed into politicized framings by media commentators or political figures (Ball, 2021).

Academic messaging could reach members of the public either directly, through individuals who follow academics on Twitter, or indirectly, through an information chain in which journalists, policymakers, and other intermediaries encounter and relay academic content to other audiences. Several pathways may help explain behavioral responsiveness to this kind of communication. One is the role of social trust as a heuristic for judgment under uncertainty. When individuals lack the knowledge to directly assess the risks and benefits of a hazard, their judgments tend to be guided by trust in the experts managing that hazard (Siegrist and Cvetkovich, 2000). A novel infectious disease is precisely this kind of situation, and academics who communicate publicly about the pandemic may function as trusted sources whose assessments help audiences interpret risks that are difficult to evaluate independently. In addition, the credibility of the source also matters. Research on persuasion and health communication has found that sources perceived as both trustworthy and expert are more effective at changing behavior, and that enlisting credible voices can make public health messages more impactful during epidemics (Bavel et al., 2020). To the extent that academics are perceived as credible sources, their messaging may carry weight beyond its informational content alone. A third plausible pathway involves the formation of parasocial bonds. Utz et al. (2022) found that sustained exposure to scientists during the pandemic was associated with stronger parasocial phenomena, and that these phenomena mediated the relationship

between exposure and behavioral outcomes such as preventive behavior. This suggests that academics who maintained a visible presence on social media may have influenced behavior not only through the information they provided but also through the trust and familiarity that accumulated over repeated exposure.

It is also possible, however, that the relationship between academic messaging and behavior operates less through any single direct or indirect channel and more through what academic communication signals about the broader pandemic information environment. Academics who closely monitor emerging research and epidemiological data may shift more of their public communication toward COVID-related science during periods when scientific, media, policy, and public attention are also elevated. In these cases, the scientific messaging share functions less as an independent input into behavior and more as an indicator of a wider information environment associated with behavioral adjustment. These interpretations are not necessarily mutually exclusive. Trust, credibility, and parasocial bonds may help explain why expert communication matters when it reaches audiences, while the information-environment interpretation helps explain why academic communication may track behavioral change even when direct audience exposure is uncertain. Our empirical design does not distinguish between these pathways, but both are consistent with the temporal associations we examine.

2.2 Communication, human mobility, and public health during COVID-19

Researchers benefited from standardized datasets that emerged during the pandemic. An example of which is population-level mobility data, which became more accessible through smartphone applications and location-based services (Wang et al., 2021). Companies such as Google, Apple, Baidu, and Tencent released aggregate mobility metrics showing how populations responded to restrictions over time. Google’s mobility dataset (Aktay et al., 2020), for instance, is generated from aggregated and anonymized user location histories and represents daily comparisons to pre-pandemic movement levels (Aktay et al., 2020). Cross-national pandemic indicators were also made available through the *Our World in Data* initiative, including the Oxford COVID-19 stringency index, which measures the strictness of government policies across multiple countries (Hale et al., 2021; Roser, 2021). The index is a composite measure of nine policy responses, including workplace and school closures, travel restrictions, limitations on public gatherings, and public information campaigns.

Multiple studies found that stringency measures such as stay-at-home orders and workplace closures are associated with reductions in human mobility (Li et al., 2021; Maloney et al., 2023). Researchers also examined how these policies and mobility changes were associated with epidemiological outcomes, such as decreases in COVID-19 cases and lower mortality rates (Agyapon-Ntra and McSharry, 2023; Gao et al., 2020; Li et al., 2021; Maloney et al., 2023; Oh et al., 2021; Wang et al., 2020; Xiong et al., 2020; Zhang et al., 2022). While many studies focus on individual national contexts (Gao et al., 2020; Li et al., 2021; Maloney et al., 2023; Wang et al., 2020), the cross-national scope of available data has also allowed analyses across countries (Agyapon-Ntra and McSharry, 2023; Oh et al., 2021; Wang et al., 2021).

The effectiveness of public communication as part of these policies varied across contexts.

Public information campaigns, which form part of the Oxford stringency index, showed mixed results. For example, [Li et al. \(2021\)](#) found that in the United States, these campaigns were associated with a reduction in the growth rate of confirmed COVID-19 cases but had no significant associations with mobility patterns or mortality rates. The cross-country analysis by [Agyapon-Ntra and McSharry \(2023\)](#) concluded that, compared to other stringent policy measures, public information campaigns had minimal associations with reductions in COVID-19 cases.

It is worth noting that communication efforts include not only official government announcements but also the presentation of science in traditional and digital media. Evidence suggests that the scientific content and tone of media coverage is associated with public mobility behavior. For example, in the UK, it was revealed that mobility changes correlated not simply with the amount of COVID-19 news coverage, but with how news emphasized scientific evidence related to public health measures ([Chan et al., 2023](#)). When news articles clearly communicated the scientific rationale behind restrictions, such as lockdowns or social distancing, it appears that there was an associated reduction in recreational outings (e.g., park visits) ([Chan et al., 2023](#)).

These findings suggest that mass communication can potentially be associated with public behavior during a health crisis. However, both government-initiated campaigns and traditional news reporting involve intermediaries. Government officials decide which information is disseminated, while journalists select which expert opinions to feature. These mediation processes can introduce delays, selectivity, and potential biases into the communication flow. In short, prior research on the nexus of communication and mobility has primarily focused on top-down or one-to-many channels, where academics' voices reach the public indirectly or selectively.

What remains unclear is whether this relationship extends to unmediated communication by academics themselves. Academic Twitter cuts through these filters and offers a more direct, if narrower, channel between scientific discourse and public attention. During the pandemic, some of the most visible voices on the platform were public health researchers and physicians, and their content ranged from technical epidemiological commentary to direct guidance on protective behaviors ([Pollett and Rivers, 2020](#); [Tardy, 2023](#)). If this discourse contained information relevant to population-level behavior, we might expect it to be associated with subsequent changes in mobility, including in categories of travel where individuals have genuine behavioral flexibility. Testing this possibility requires linking daily variation in scientific communication to daily variation in human mobility across countries, while accounting for several confounders. To the best of the authors' knowledge, no prior study has done so.

2.3 Cultural factors during the pandemic

We have good reasons to expect that the association between scientific messaging and mobility varies by cultural context. Culture has been defined as "the collective programming of the mind that distinguishes the members of one group or category of people from others" ([Hofstede, 2011](#)), and these collective patterns lead people in different societies to respond to the same events in distinct ways. A growing body of work has linked cultural factors to individuals' health beliefs, behaviors, and use of healthcare services ([Zidkova et al., 2025](#)).

The human response to COVID-19 is therefore also likely to have been shaped by national culture, including how societies understood risk and accepted restrictions.

One of the most widely used frameworks for operationalizing cultural characteristics across countries is Hofstede’s cultural dimensions framework (Hofstede, 2011). The framework was originally built around four dimensions. Power distance captures the degree to which less powerful members of a society accept that power is distributed unequally, with high values reflecting acceptance of social hierarchy and low values reflecting more egalitarian relationships. Individualism captures the degree to which people see themselves as independent rather than as part of a wider community, with high values indicating individualist orientations and low values indicating collectivist ones. Motivation towards achievement and success (formerly masculinity) captures the degree to which a society values competition, achievement, and material rewards, with high values pointing to more competitive orientations and low values pointing to a greater emphasis on cooperation and quality of life. Uncertainty avoidance captures the degree to which a society feels threatened by ambiguity and the unknown, with high values indicating a preference for clear rules and stable conditions and low values indicating greater comfort with ambiguity. The framework was later expanded to include two additional dimensions: long-term orientation, which reflects the degree to which a society prioritizes future-oriented values such as perseverance and saving, and indulgence, which reflects the degree to which a society allows the relatively free gratification of natural human desires, with low values reflecting more restraint.

Hofstede’s dimensions have been applied in cross-country empirical research related to population health (Zidkova et al., 2025; Mackenbach, 2014), including outcomes, such as obesity (Tekeş et al., 2019), body mass index (Masood et al., 2019), depression (Arrindell et al., 2003), and suicide incidence rates (Webster Rudmin et al., 2003). Other cultural frameworks have also been explored in the pandemic literature, such as Gelfand’s tightness-looseness index (Gelfand et al., 2021), Triandis and Gelfand’s cultural personal values (Triandis and Gelfand, 1998; Blut et al., 2022), and Schwartz’ (Schwartz, 1999) cultural values (Atalay and Solmazer, 2021). We focus our review on studies which used Hofstede’s dimensions, given their wide use in cross-country empirical work and their direct relevance to the moderation analysis we present in this paper. We also focus mainly on studies that analyze cross-country samples (i.e., using country-level Hofstede values), although we acknowledge that other papers use within-country individualism scores (Fu et al., 2024; Chen et al., 2021; Lu et al., 2021).

Several studies have applied Hofstede’s framework to study the cultural dimensions of pandemic-related attitudes, behaviors, and outcomes. In terms of attitudes, much of the focus has been on vaccine-related responses, including vaccine hesitancy (Lu, 2023), vaccine intention (Lamot and Kirbiš, 2024), vaccine acceptance (Leonhardt and Pezzuti, 2022), and vaccine confidence (Zheng, 2024). These studies have focused specifically on a subset of dimensions, including: individualism (Zheng, 2024; Lamot and Kirbiš, 2024; Leonhardt and Pezzuti, 2022), uncertainty avoidance (Lu et al., 2021), and power distance (Lamot and Kirbiš, 2024). Lucas et al. (2022) examined how cultural values predict individuals’ support for vaccination and quarantine mandates. Ma et al. (2022) examined whether willingness to quarantine varies with long-term orientation. Shan et al. (2024) used Twitter data to analyze cultural patterns in expressed stances on COVID-19 prevention measures. In terms of outcomes, the availability of cross-country pandemic datasets, most notably *Our World*

in Data (Roser, 2021), has enabled studies linking Hofstede’s dimensions to pandemic indicators, such as case counts, deaths, vaccination rates, and growth rates (Gokmen et al., 2021; Dheer et al., 2021; Chen and Biswas, 2023).

The behavioral aspect, which is closest to the focus of our paper, has been examined in several studies using Google mobility data. Huynh (2020) examined social distancing, proxied by reduced mobility to public places, and found that among the three dimensions considered (power distance, uncertainty avoidance, and individualism), uncertainty avoidance appeared to be the strongest predictor, with higher-uncertainty-avoidance societies exhibiting fewer public gatherings. One main limitation of this study, however, is its reliance on a single cross-sectional snapshot, thus raising the possibility that the associations may not hold when another date is chosen (Wang, 2021). Wang (2021) extended the analysis to the first wave of the pandemic and explored all six Hofstede dimensions. The results indicate that, all else equal, long-term orientation and indulgence are among the strongest predictors of mobility responses. Chen et al. (2021) specifically focused on individualism using a sample of 111 countries during the first wave and found that more individualistic societies were associated with less compliance with lockdown measures.

From this, we identified several gaps. First, several papers make use of cross-sectional designs or pooled OLS specifications, of which the general idea is to combine cultural measures (time-invariant) and mobility (time-varying) into a single model. One limitation is that these specifications do not exploit within-country variation over time and are therefore limited in their ability to account for unobserved country-level heterogeneity. Pooled OLS may also raise concerns about serial correlation if country-level clustering or comparable corrections are not applied, given that daily mobility is highly autocorrelated within countries.

Second, many existing studies typically rely on contemporaneous specifications that link policy and mobility at the same point in time. This may be restrictive in the sense that behavioral responses to policy measures, public communication, or changes in the epidemiological situation are unlikely to be instantaneous. Such responses may instead unfold over several days or weeks, which makes temporal lags worth exploring.

Third, some studies do not incorporate time fixed effects, which may be a concern in a global pandemic setting. Major announcements, waves of international media attention, vaccine developments, and coordinated policy responses could influence mobility across countries at the same time. Without accounting for such common shocks, estimates may be biased due to unobserved confounding.

Finally, some studies operationalize social distancing using composite indices derived from Google mobility data. The challenge with this aggregation, on top of the aggregation already inherent in the source data, is that it may not capture heterogeneity across mobility categories. Different types of mobility may respond differently to communication and to underlying cultural predispositions; for example, workplace mobility may reflect institutional and economic constraints, while parks or retail mobility may involve greater individual discretion.

These motivate the design choices in this paper, which combines panel methods, time fixed effects, lagged specifications, and category-disaggregated mobility analysis.

3 Research questions

Building on the gaps identified and the broader role of academic and scientific communication during the pandemic, we address two research questions (see Figure 1).

RQ1. Is the daily share of academics’ Twitter activity devoted to COVID-related scientific messaging associated with subsequent changes in population mobility across countries during the early phase of the pandemic?

RQ2. Does the strength of this association depend on cultural context, as measured by Hofstede’s six cultural dimensions?

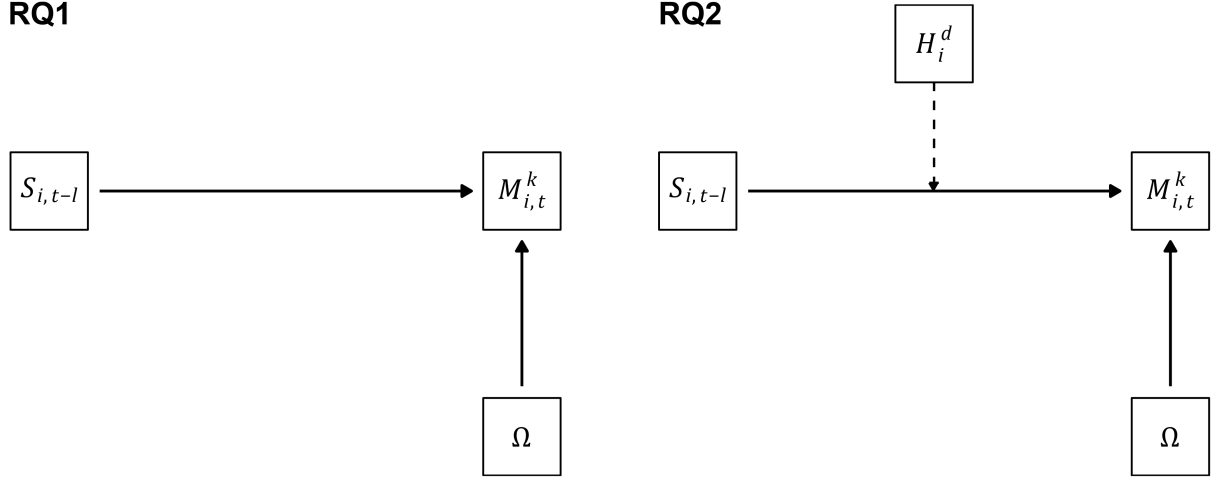


Figure 1: Conceptual diagram of the two research questions

Notes: $S_{i,t-l}$ is the share of academics’ tweets classified as scientific messaging related to COVID-19 in country i on day t , lagged by l days. $M_{i,t}^k$ is the mobility index for country i on day t for mobility type k . H_i^d is the score for country i on Hofstede cultural dimension d . Ω denotes the set of controls. Full descriptions of the variables are listed in Equation 1 and Equation 2.

4 Data and methods

4.1 Data collection and processing

We relied on Twitter data in our analyses. We identified academics using a snowball sampling approach, starting with existing Twitter lists of academics covering various disciplines. We then used these users as seed accounts and downloaded the accounts they followed. We verified whether the followed users were academics and applied the same approach to newly identified users. Through this iterative snowball procedure, we downloaded the accounts of all identified academics, followed by those of professors and senior academics, from January to August 2022. This resulted in the identification of 504,288 professors and senior academics

(see Supplementary Material). We limited our sample to professors and senior academics, as they were easier to identify, and the classification yielded few false-positive results. Overall, in a manual validation of a random sample of 318 accounts, we achieved a precision of 0.962.

Using Twitter’s Academic API, we retrieved in November 2022 the last 3,200 tweets published by each user. This means we have complete data for 500,387 academics, which is approximately 99% of the total sample. Only a small number of users exceeded the API-imposed limit of 3,200 tweets, meaning that their downloaded tweets do not completely cover the time range of our analysis in 2020. This provided us with a dataset of over 500 million tweets, more than 50% of which are English tweets, based on the *lang* field of the metadata.

We used the *location* field provided in the tweets’ metadata to identify the country from which the tweets were posted. We found that 385,201 users had information in their *location* fields. The countries corresponding to the provided locations were identified by directly accessing the OpenAI API from within R and using the *gpt-4o-mini-2024-07-18* model with a temperature of 0. No pre-processing was applied to the location texts. We used the following prompt for country identification:

“Your task is to determine which country the given place name is located in. If the name corresponds to an actual location (this can include a full country name, common abbreviation like ‘D’ for Germany, IATA airport codes also count as location, or an emoji like [Qatar flag emoji]¹ for Qatar), return only the full country name (e.g., ‘Germany’, ‘Qatar’). If there is more than one location mentioned, use the first one. If it does not represent a valid location, return ‘NA’. Here is the name of the place: [information from location field].”

The following system role was used: “You are a helpful assistant identifying what country a given place name is located.”

We then post-processed the identification results and manually cleaned systematic errors. The results were validated through a random sample of 500 users with identified locations and another random sample of 500 users whose locations were not identified (NA cases) by the method previously described. Manual validation revealed an accuracy of 99.60% for users with identified locations and 69.60% for users without identified locations.

Because the accuracy for users without identified locations was too low, we used users’ profile descriptions to further infer whether their locations could be identified. After the initial identification, 137,300 users remained without an identified country of origin. We again accessed OpenAI API, using a different prompt tailored to profile descriptions (see Supplementary Methods for the prompt and system role). For this second identification, we also did not pre-process the description texts. This round of identification was manually validated using a random sample of 500 users from the remaining cases, achieving an accuracy of 95%. In total, we were able to identify the locations of 409,335 users using the two identification approaches.

For our analysis, we selected only the English tweets from February 15 to December 31, 2020, resulting in 28,985,603 English tweets from 244,132 unique users. For this corpus, only 233,186 unique users (17,195,240 tweets) tweeted with an identified location.

¹The original prompt includes the Unicode flag emoji, which is rendered here in text form for compatibility with typesetting.

4.2 Identifying scientific messaging in tweets

To examine the scientific communication of academics on Twitter during the COVID-19 pandemic, we first identified English-language tweets from 2020 to 2021 that were related to COVID-19 using a list of keywords (e.g., COVID, COVID-19, coronavirus, pandemic; see Supplementary Methods for the full list). This dictionary-based automatic classification was manually validated using a random sample of 200 tweets, resulting in a Cohen’s κ of 0.928 and an accuracy of 99%.

We then developed a coding guideline and delineated the scope of our definition of science communication (Bichara et al., 2022; Chan et al., 2023; Guenther et al., 2023; Mach et al., 2021). Two manual coders independently classified a random sample of 100 tweets according to whether they communicated scientific knowledge about COVID-19 based on the agreed-upon guideline (see Supplementary Methods). We compared the inter-rater agreement between the two coders, yielding a Cohen’s κ of 0.88. After establishing that we can rely on our definition of science communication regarding COVID-19, we accessed the OpenAI API from within R to automatically classify the tweets. We developed several prompts and evaluated them by classifying the already labeled random sample of 100 tweets according to our defined guidelines. We then selected the prompt that produced the best performance, with a Cohen’s κ of 0.70, with an accuracy of 85% for the random sample. We used the ChatGPT model *gpt-4o-mini-2024-07-18*, with a temperature of 0, for the classification. We did not pre-process the texts to avoid modifying the semantics of the tweets.

We used the following system role: "You are a helpful assistant for identifying the relevance of a tweet to science communication during the COVID-19 pandemic."

Meanwhile, for the selected prompt, we used the following: "Classify whether a tweet is a form of science communication regarding COVID-19. Relevant tweet to science communication of COVID-19 directly discusses, shares, or critiques scientific findings, research, or expert consensus related to COVID-19 (e.g., virology, epidemiology, vaccines, public health measures). Non-relevant tweet to science communication of COVID-19 does not focus on scientific communication of knowledge on COVID-19 (e.g., personal opinions, political debates, non-scientific news, memes). Return only a single number: 0) Not Relevant, 1) Relevant. The tweet is: [tweet]"

We then manually validated the classified tweets using a new random sample of 500 tweets, which resulted in a Cohen’s κ of 0.83 and an accuracy of 92%. We provide paraphrased examples of tweets classified as scientific and non-scientific COVID-related messaging in the Supplementary Material.

4.3 Analytical strategy

For RQ1, we estimate a set of panel linear regression models that relate academics’ scientific messaging on COVID-19 to subsequent population mobility (see Equation 1).

$$M_{i,t}^k = \alpha_i^k + \beta_1^k M_{i,t-1}^k + \beta_2^k S_{i,t-l} + \beta_3^k T_{i,t-l} + \beta_4^k P_{i,t-l} + \beta_5^k \log(C_{i,t-l}) + \beta_6^k \log(D_{i,t-l}) + \gamma_w^k + \varepsilon_{i,t}^k. \quad (1)$$

where:

- $M_{i,t}^k$ is the mobility index for country i on day t for mobility type k ;

- α_i^k denotes country fixed effects for mobility type k ;
- $M_{i,t-1}^k$ is the one-day lagged dependent variable;
- $S_{i,t-l}$ is the 7-day rolling average of the share of academics' tweets classified as scientific messaging related to COVID-19 in country i , lagged by l days;
- $T_{i,t-l}$ is the 7-day rolling average of tweets per academic user in country i , lagged by l days;
- $P_{i,t-l}$ is the policy stringency index in country i on day t , lagged by l days;
- $C_{i,t-l}$ is the 7-day rolling average of daily new COVID-19 cases in country i , lagged by l days;
- $D_{i,t-l}$ is the 7-day rolling average of daily new COVID-19 deaths in country i , lagged by l days;
- γ_w^k denotes week fixed effects;
- $\varepsilon_{i,t}^k$ is the error term.

The dependent variable, $M_{i,t}^k$, is the mobility index for country i on day t for mobility type k . Values are taken from Google's COVID-19 Community Mobility Reports (Aktay et al., 2020). We estimate a separate model for each of the six mobility types k (retail and recreation, grocery and pharmacy, parks, transit stations, workplaces, and residential).

The main independent variable of interest, $S_{i,t-l}$, is the scientific messaging share, which we operationalize as the 7-day rolling average of the percentage of academics' tweets in country i that are classified as scientific messaging about the pandemic, lagged by l days. Conceptually, this measure is intended to capture the prominence of COVID-related scientific communication within academics' broader public discourse. We use a 7-day rolling average to smooth day-to-day volatility and to reflect sustained short-term patterns in academic communication.

We consider three lag lengths (1, 7, and 14 days) estimated in separate models, which allows us to capture how any association may occur across different time horizons. Taken together with the six mobility types, this results in 18 main models for RQ1 (6 mobility types $\times$ 3 lag lengths).

The remaining terms are controls. The one-day lagged dependent variable, $M_{i,t-1}^k$, captures persistence in mobility patterns, since mobility on a given day is typically a strong predictor of mobility the following day. We acknowledge that including a lagged dependent variable in a fixed-effects model may introduce Nickell bias, which is of order $1/T$ (Nickell, 1981). However, because our panel has a relatively long dimension, with approximately 301 daily observations per country after accounting for rolling averages and lag structures (see Section 4.4), this bias is likely to be limited and unlikely to affect the interpretation of the results. $P_{i,t-l}$ is the Oxford policy stringency index for country i , which we included to adjust for the formal policy environment. The epidemiological situation is represented by $C_{i,t-l}$ and $D_{i,t-l}$, which are the 7-day rolling averages of daily new COVID-19 cases and deaths, respectively. $T_{i,t-l}$ is the 7-day rolling average of tweets per academic user in country i ,

which adjusts for the overall levels of academic Twitter activity. The controls are lagged by the same l days as the main predictor, with the exception of the one-day lagged dependent variable.

α_i^k absorbs time-invariant differences across countries, while γ_w^k absorbs global common shocks such as WHO announcements or coordinated media events. $\varepsilon_{i,t}^k$ is the residual.

For RQ2, we extend the model in Equation 1 to examine whether any association between scientific messaging and mobility varies across cultural contexts. We introduce an interaction term between the scientific messaging share and each of Hofstede’s six cultural dimensions, as shown in Equation 2.

$$M_{i,t}^k = \alpha_i^k + \beta_1^k M_{i,t-1}^k + \beta_2^k S_{i,t-l} + \beta_3^k T_{i,t-l} + \beta_4^k P_{i,t-l} + \beta_5^k \log(C_{i,t-l}) + \beta_6^k \log(D_{i,t-l}) + \beta_7^k (S_{i,t-l} \times H_i^d) + \gamma_w^k + \varepsilon_{i,t}^k. \quad (2)$$

where:

- H_i^d is the score for country i on Hofstede cultural dimension d

Note that H_i^d does not enter the model as a standalone term because it is time-invariant within country and is therefore absorbed by the country fixed effect α_i^k . The interaction term remains identifiable since it varies over time through $S_{i,t-l}$. We estimate a separate model for each combination of mobility type k , lag length l , and cultural dimension d , which results in 108 main models for RQ2 (6 mobility types $\times$ 3 lag lengths $\times$ 6 cultural dimensions). The values for the cultural dimensions are standardized prior to estimations.

Models are estimated using the within (fixed-effects) estimator, which demeans all variables by their country-specific means and thereby absorbs time-invariant cross-country heterogeneity. Standard errors are clustered at the country level using the cluster-robust variance estimator in `plm::vcovHC` (with `cluster = "group"`), which accounts for within-country serial correlation and heteroskedasticity. Estimations are conducted in R.

We also perform several robustness checks to assess the sensitivity of our estimates. First, we re-estimate the main models using alternative thresholds for the minimum daily tweet volume by academics per country, as described in Section 4.4. Second, we restrict the sample to non-English-speaking countries to assess whether the associations hold beyond the predominantly English-speaking subset. Finally, we conduct a leave-one-out analysis in which each country is excluded in turn to verify that no single country disproportionately drives the main results.

4.4 Analytical sample

The final analytical sample consists of daily country-level observations from February 15 to December 31, 2020. This period corresponds to the pre-vaccination phase of the pandemic, when public health responses continued to rely on non-pharmaceutical interventions and behavioral adjustment.

We constructed the analytical panel by first identifying the countries represented in our sample of academics on Twitter. We then retained countries that could be matched to Google Community Mobility Reports, epidemiological indicators, and policy stringency data.

Because our main independent variable is measured as the daily percentage of academics’ tweets containing COVID-related scientific content, we restricted the sample to countries with at least 50 tweets from academics on each day of the study period. This threshold was adopted to improve the stability of the aggregate percentage measure used in the country-level models. The resulting initial sample includes 42 countries observed over 321 days, yielding 13,482 country-day observations.

As a robustness check, we also re-estimated the analyses using alternative minimum daily tweet cutoffs of 25 and 100 tweets. The main patterns and interpretations remain unchanged across these specifications. We provide the details in the Supplementary Material.

For compatibility across models, we use a common estimation sample with complete observations for the dependent and independent variables. The 7-day rolling averages for some of the independent variables, combined with the 14-day lag structure, remove the first 20 days for each country from the estimation sample, corresponding to 840 country-day observations. A small number of additional observations are excluded because of missing grocery and pharmacy mobility values: three in Norway, one in Poland, and one in Switzerland. Since these missing values affect both the observed outcome ($M_{i,t}^k$) and the 1-day lagged mobility control ($M_{i,t-1}^k$), they account for 10 further excluded observations. The resulting panel for RQ1 is therefore slightly unbalanced, with $N_c = 42$ countries and $N_{obs} = 12,632$ valid country-day observations.

For RQ2, we retain the largest possible estimation sample for each cultural dimension, reducing the sample only when Hofstede scores are unavailable. Uganda has no available Hofstede values for the dimensions used in the main moderation analysis. Thus, for PDI, IDV, MAS, UAI, and LTO, the analytical sample includes $N_c = 41$ countries and $N_{obs} = 12,331$ country-day observations. This corresponds to the RQ1 sample of 12,632 observations minus 301 observations from Uganda.

For IVR, Hofstede values are also unavailable for Israel and Kenya. The analytical sample for IVR is therefore further reduced to $N_c = 39$ countries and $N_{obs} = 11,729$ country-day observations (i.e., exclusion of Uganda, Israel, and Kenya from the RQ1 estimation sample).

We present the descriptive statistics in Table 1.

5 Results

5.1 Descriptives

We begin by presenting the patterns of the dependent variable. Figure 2 shows how the six mobility types changed across countries over the study period, with values expressed as percentage changes relative to a pre-pandemic baseline. There was a noticeable decline across out-of-home mobility categories in the early weeks, accompanied by an increase in residential mobility, as populations moved less and stayed home more. Countries such as Italy, Spain, Ireland, and the UK recorded early declines of up to 60 to 80% in several out-of-home categories.

In much of Western and Northern Europe (e.g., Austria, France, Germany, the Netherlands, Ireland, the UK, and the Nordic countries), park mobility increased through the summer months, reaching levels well above the pre-pandemic baseline before declining again in autumn. This is less evident, or absent altogether, in countries in the Gulf (e.g., Saudi Ara-

Table 1: Descriptive statistics for main model variables.

Variable	Mean	SD	Min	Median	Max
Dependent variables (% change from baseline)					
Retail & Recreation	-29.01	24.07	-96.00	-23.00	54.00
Grocery & Pharmacy	-7.61	18.82	-94.00	-4.00	162.00
Parks	7.98	64.75	-95.00	-9.00	517.00
Transit Stations	-34.82	21.43	-92.00	-35.00	38.00
Workplaces	-26.28	19.55	-92.00	-24.00	32.00
Residential	11.62	8.53	-13.00	10.00	55.00
Main independent variable					
Scientific messaging share $S_{i,t}$ (%)	4.98	3.57	0.14	3.72	27.97
Controls					
Tweets per academic $T_{i,t}$	0.31	0.22	0.03	0.26	4.95
Policy stringency $P_{i,t}$	64.13	17.94	0.00	66.20	100.00
Daily new cases $C_{i,t}$ (log)	6.22	2.46	0.00	6.27	12.29
Daily new deaths $D_{i,t}$ (log)	2.69	2.10	0.00	2.33	7.93
Cultural dimensions (Hofstede)					
Power Distance (PDI)	53.46	21.06	11.00	57.00	94.00
Individualism (IDV)	51.98	26.08	0.00	56.00	100.00
Achievement/Success (MAS)	51.54	19.24	5.00	56.00	95.00
Uncertainty Avoidance (UAI)	64.49	22.23	8.00	65.00	100.00
Long-Term Orientation (LTO)	43.49	22.51	1.00	47.00	100.00
Indulgence (IVR)	55.44	21.99	0.00	59.00	100.00

Note: Country-day variables are based on the RQ1 estimation sample ($N = 12,632$ country-day observations). $S_{i,t}$, $T_{i,t}$, $C_{i,t}$, and $D_{i,t}$ are 7-day rolling averages. Descriptive statistics for Hofstede dimensions are reported at the country level, with one observation per country ($N = 41$ for PDI, IDV, MAS, UAI, and LTO; $N = 39$ for IVR). Hofstede dimensions are standardized in the regression models.

bia, UAE), parts of Africa (e.g., Kenya, Nigeria, Uganda), Southeast Asia (e.g., Singapore, the Philippines), and parts of Latin America (e.g., Mexico, Venezuela, Chile, Colombia), where most mobility categories other than residential remained below baseline for much of the study window. The exceptions include countries such as New Zealand, where mobility appeared to return close to pre-pandemic levels by late summer after a sharp initial drop.

In terms of the main independent variable, Figure 3 shows the corresponding trends in academics’ scientific messaging about COVID-19. In general, the share of tweets classified as scientific messaging related to the pandemic rose sharply in the early weeks of the pandemic, typically peaking around March or April 2020, before gradually declining through the summer and into the autumn. By late 2020, most countries had settled to substantially lower levels.

5.2 RQ1: Scientific messaging and population mobility

We then examine the associations between the scientific messaging share and population mobility (RQ1). Figure 4 plots the coefficient of the lagged scientific messaging share across the 18 main models, with 95% confidence intervals based on country-clustered standard errors.

The overall pattern suggests that the scientific messaging share is negatively associated

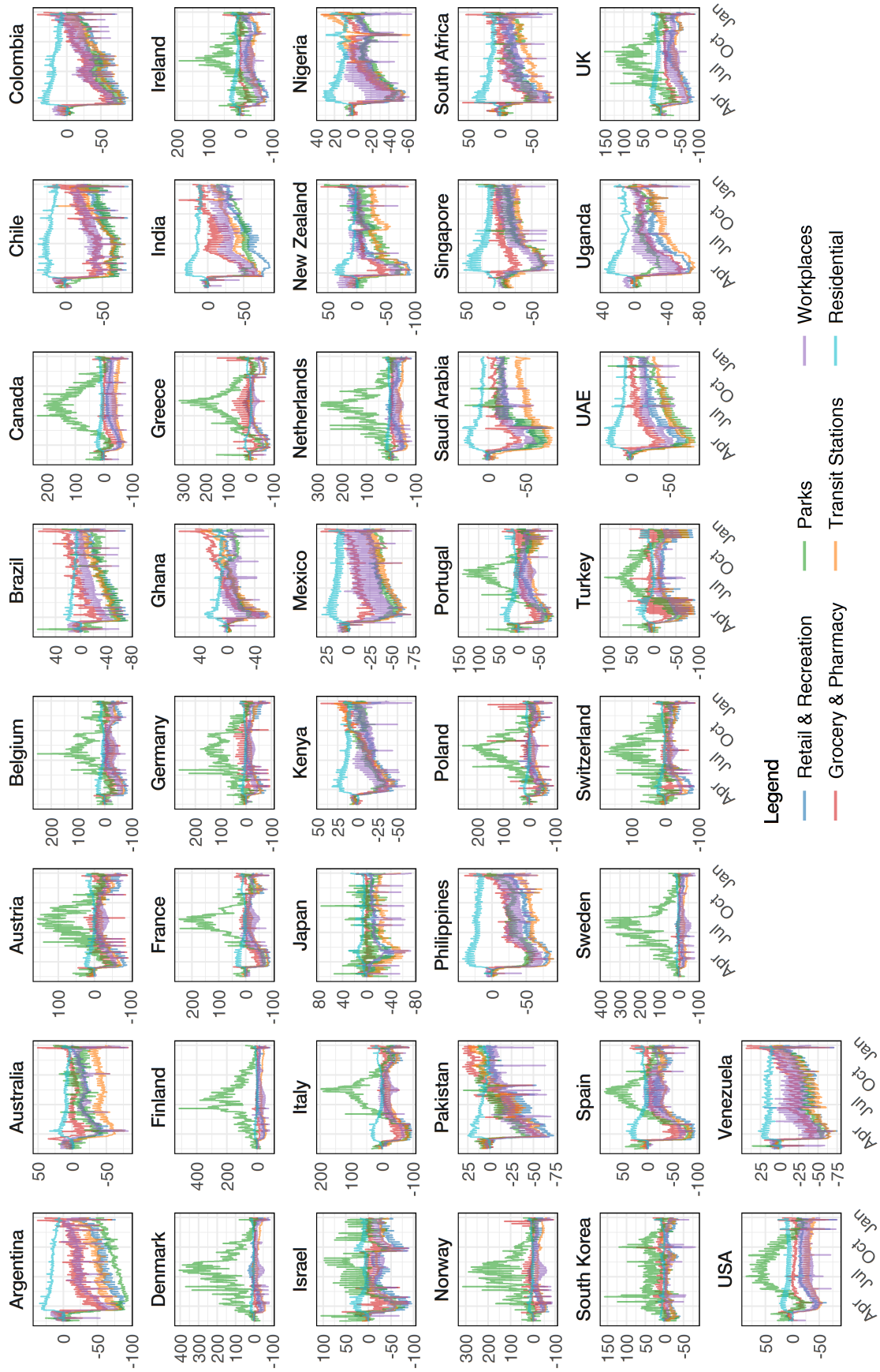


Figure 2: Trends in human mobility during COVID-19 across countries

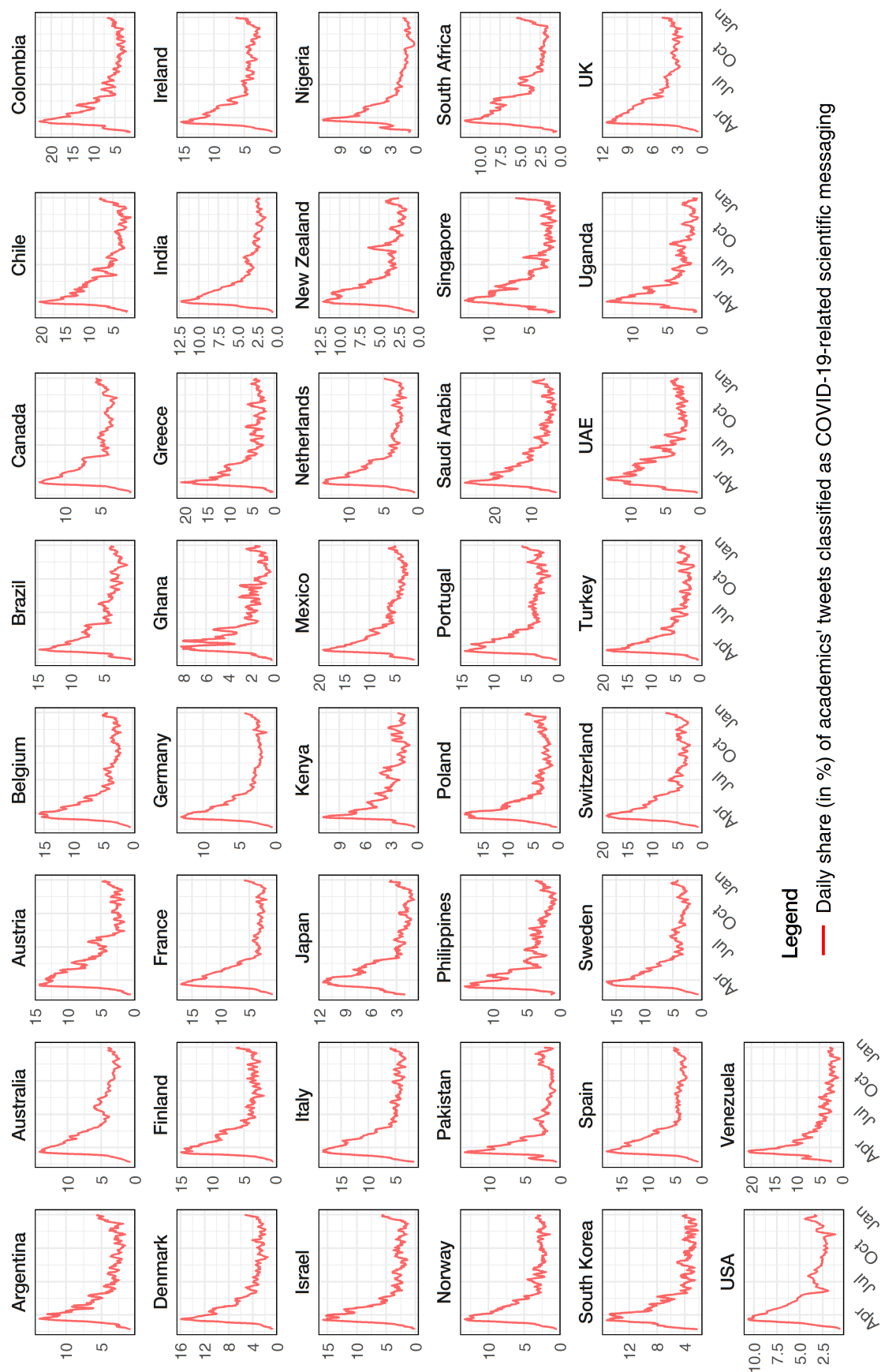


Figure 3: Trends in academics' COVID-19-related scientific messaging across countries

Note: Values are expressed as 7-day rolling averages.

with out-of-home mobility and positively associated with residential mobility, *ceteris paribus*. This is in line with the expectation that populations spend less time in public settings and more time at home following days when a larger share of academics' Twitter activity is devoted to scientific communication about COVID-19.

Workplace mobility shows the most consistent association with the scientific messaging share. The coefficients are negative and statistically significant across all three lag lengths: 1 day ($\beta = -0.840$, 95% CI $[-1.269, -0.411]$, $p < 0.001$), 7 days ($\beta = -0.691$, 95% CI $[-1.050, -0.333]$, $p < 0.001$), and 14 days ($\beta = -0.380$, 95% CI $[-0.695, -0.064]$, $p = 0.018$). Workplace mobility is also the only category for which the association remains statistically significant at the 14-day lag.

We also find associations with more discretionary forms of out-of-home mobility, though these are concentrated at shorter lags. The scientific messaging share is negatively associated with retail and recreation mobility at the 1-day lag ($\beta = -0.457$, 95% CI $[-0.707, -0.207]$, $p < 0.001$) and 7-day lag ($\beta = -0.293$, 95% CI $[-0.518, -0.068]$, $p = 0.011$). A similar pattern is observed for parks mobility, with significant negative associations at the 1-day lag ($\beta = -0.568$, 95% CI $[-1.006, -0.129]$, $p = 0.011$) and 7-day lag ($\beta = -0.408$, 95% CI $[-0.798, -0.018]$, $p = 0.040$). Residential mobility shows the corresponding positive pattern at the same lag lengths: 1 day ($\beta = 0.364$, 95% CI $[0.218, 0.511]$, $p < 0.001$) and 7 days ($\beta = 0.250$, 95% CI $[0.134, 0.365]$, $p < 0.001$).

The association with transit station mobility is only observed at the 1-day lag ($\beta = -0.350$, 95% CI $[-0.589, -0.110]$, $p = 0.004$). We find no statistically significant associations for grocery and pharmacy mobility at any of the three lag lengths.

5.3 RQ2: Variation across cultural dimensions

We now examine whether the magnitude and significance of these associations vary across cultural contexts. Figure 5 plots the coefficient of the interaction term between the lagged scientific messaging share and each of the six Hofstede cultural dimensions across the 108 main models for RQ2. A coefficient significantly different from zero indicates that the messaging-mobility association varies across countries with different scores on a given cultural dimension.

In countries scoring higher on power distance, the negative association between the scientific messaging share and out-of-home mobility is more pronounced. This is observed for grocery and pharmacy, parks, and transit station mobility at all three lag lengths, for retail and recreation at the 7- and 14-day lags, and for workplaces at the 14-day lag. The positive association with residential mobility is also stronger at the 7- and 14-day lags. A similar pattern is observed for uncertainty avoidance. In high-UAI countries, the negative association is stronger for retail and recreation (1-day lag), grocery and pharmacy (1-day lag), parks (all lags), and transit stations (1- and 7-day lags), with the positive association with residential mobility also stronger at the 1-day lag. These are in line with theoretical expectations, as cultures inclined to accept authority (high PDI) and to seek stability in situations of ambiguity (high UAI) may be more receptive to expert guidance when uncertainty is high, such as during the early phase of a pandemic.

The association is also more pronounced in countries scoring higher on motivation towards achievement and success, particularly for retail and recreation mobility (all lags) and transit

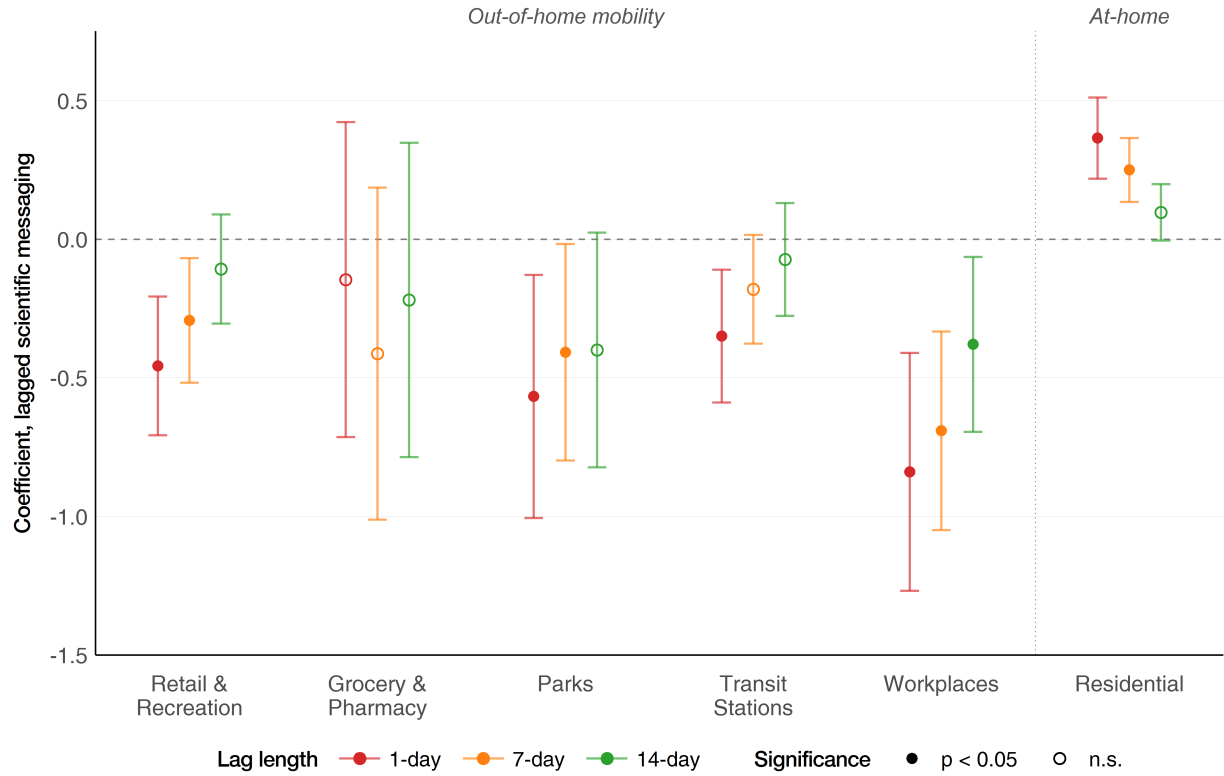


Figure 4: Associations between scientific communication by academics and subsequent population mobility, by trip type and temporal lag

Note: Points show coefficients from panel linear regressions of each Google Mobility category on lagged scientific messaging (1, 7, and 14 days), with country and week fixed effects. Models adjust for lagged mobility, epidemiological conditions, policy stringency, and academic tweeting intensity. Error bars are 95% confidence intervals with country-clustered standard errors.

station mobility (7- and 14-day lags).

The significant interactions with long-term orientation operate in the opposite direction. In contrast to the strengthening pattern observed for PDI, UAI, and MAS, higher LTO scores consistently attenuate the association between the scientific messaging share and mobility across all mobility types and lag lengths. Individualism shows a similar but more selective pattern, with higher IDV scores attenuating the association for grocery and pharmacy mobility and transit station mobility across all lag lengths.

Interactions for indulgence are smaller in magnitude and not statistically significant.

Across most mobility categories, cultural moderation is observed, though not uniformly across dimensions or lag lengths. Transit station mobility is moderated by the largest set of dimensions (PDI, IDV, MAS, UAI, and LTO at various lags). Grocery and pharmacy mobility is moderated by four dimensions: PDI, IDV, UAI, and LTO. Retail and recreation mobility is also moderated by four dimensions: PDI, MAS, UAI, and LTO. Parks and residential mobility are each moderated by PDI, UAI, and LTO at certain lag lengths. Workplace mobility is moderated to a lesser extent, with significant interactions observed only for PDI and LTO.

5.4 Robustness checks

We assess the sensitivity of our main findings across three aspects. First, we re-estimate the models using alternative thresholds for the minimum daily tweet volume by academics per country. Second, we restrict the sample to non-English-speaking countries to verify that the associations hold beyond the English-speaking subset. Across both checks, the direction and magnitude of the main estimates are consistent with the full-sample results, though some borderline models (those where zero falls just inside or outside the 95% confidence interval) shift in or out of conventional significance. Full results are provided in the Supplementary Material.

Third, we conduct a leave-one-out sensitivity analysis to assess whether any single country disproportionately drives the main findings. For each of the 18 RQ1 models, we re-estimate the specification 42 times, each time excluding one country (see Table 2). In general, the results are stable across all leave-one-out runs, with significant and non-significant models retaining their status in nearly all cases. We nonetheless observe a small number of borderline models (e.g., Parks, 14-day; Transit, 7-day) which show occasional significance in individual leave-one-out runs, which is expected given that their full-sample estimates are near the conventional significance threshold.

6 Conclusions

This study examined whether the daily share of academics' Twitter activity devoted to COVID-related scientific messaging was associated with subsequent shifts in population mobility across 42 countries during the early phase of the pandemic, and whether these associations varied across cultural contexts. We found that a higher scientific messaging share was associated with reductions in several out-of-home mobility categories and increases in residential mobility. The association was most consistent for workplace mobility, present at shorter lags for transit and residential categories, and emerged only at longer lags for

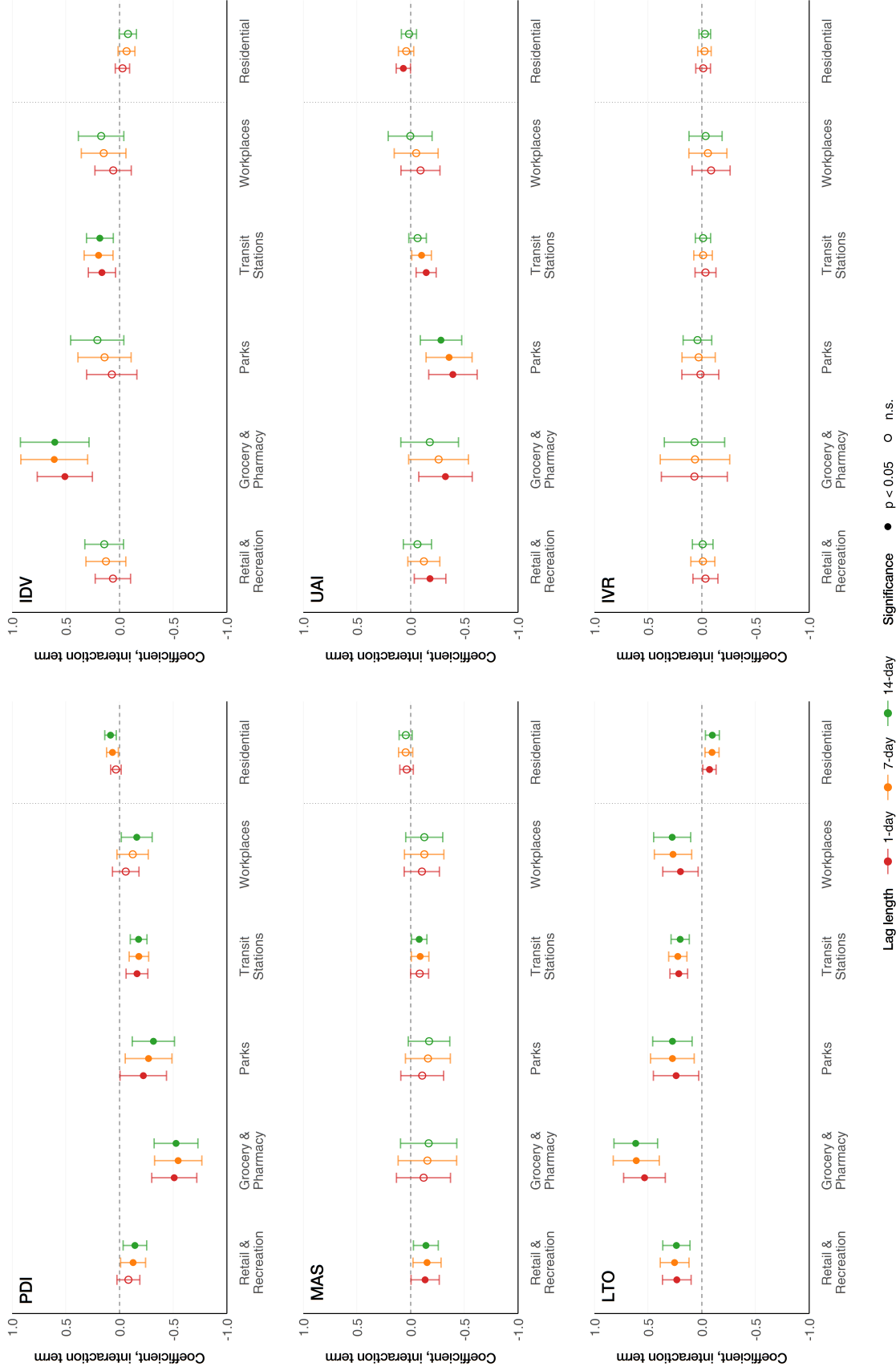


Figure 5: Interaction between scientific messaging by academics and Hofstede cultural dimensions as predictors of population mobility, by trip type and temporal lag.

Note: Each panel shows the estimated coefficient on the interaction term between lagged scientific messaging and one of Hofstede's six cultural dimensions: power distance (PDI), individualism (IDV), motivation towards achievement and success (MAS), uncertainty avoidance (UAI), long-term orientation (LTO), and indulgence (IVR). Error bars are 95% confidence intervals with country-clustered standard errors.

Table 2: Leave-one-out sensitivity analysis for RQ1 models.

Model	Full β	Full 95% CI	LOO range	Sig. (of 42)
<i>Significant in full sample</i>				
Parks, 1-day	-0.568	[-1.006, -0.129]	[-0.678, -0.447]	42/42
Parks, 7-day	-0.408	[-0.798, -0.018]	[-0.530, -0.282]	34/42
Residential, 1-day	0.364	[0.218, 0.511]	[0.327, 0.423]	42/42
Residential, 7-day	0.250	[0.134, 0.365]	[0.218, 0.281]	42/42
Retail & Recreation, 1-day	-0.457	[-0.707, -0.207]	[-0.546, -0.392]	42/42
Retail & Recreation, 7-day	-0.293	[-0.518, -0.068]	[-0.357, -0.240]	42/42
Transit Stations, 1-day	-0.350	[-0.589, -0.110]	[-0.431, -0.306]	42/42
Workplaces, 1-day	-0.840	[-1.269, -0.411]	[-1.040, -0.776]	42/42
Workplaces, 14-day	-0.380	[-0.695, -0.064]	[-0.463, -0.298]	42/42
Workplaces, 7-day	-0.691	[-1.050, -0.333]	[-0.835, -0.617]	42/42
<i>Not significant in full sample</i>				
Grocery & Pharmacy, 1-day	-0.146	[-0.714, 0.422]	[-0.280, -0.043]	0/42
Grocery & Pharmacy, 14-day	-0.219	[-0.786, 0.347]	[-0.376, -0.111]	0/42
Grocery & Pharmacy, 7-day	-0.413	[-1.012, 0.186]	[-0.570, -0.295]	0/42
Parks, 14-day	-0.400	[-0.823, 0.023]	[-0.562, -0.269]	5/42
Residential, 14-day	0.096	[-0.005, 0.198]	[0.070, 0.138]	8/42
Retail & Recreation, 14-day	-0.108	[-0.305, 0.089]	[-0.191, -0.062]	0/42
Transit Stations, 14-day	-0.073	[-0.277, 0.130]	[-0.158, -0.035]	0/42
Transit Stations, 7-day	-0.181	[-0.377, 0.015]	[-0.256, -0.147]	5/42

Notes: Each model is re-estimated 42 times, each time excluding one country. Full β and 95% CI are from the full-sample specification. LOO range reports the minimum and maximum coefficient estimates across the 42 leave-one-out runs. Sig. (of 42) reports how many of the 42 runs produce a coefficient significant at $p < 0.05$.

parks. Cultural context also mattered, as the association was generally stronger in countries scoring higher on power distance and uncertainty avoidance, weaker in countries with higher long-term orientation, and showed more selective moderation by individualism and motivation towards achievement and success. Taken together, these findings suggest that scientific communication by academics was embedded in a broader information environment associated with behavioral adjustment during the pandemic, and that its relationship with population mobility was not uniform across societies.

We note, however, that the associations we observe should not be interpreted as evidence that tweets by academics directly changed population behavior. Rather, we view them as indicating that academics' public communication was embedded in broader processes of collective response and information flow. On days when academics devoted a larger share of their Twitter activity to COVID-related scientific messaging, traditional media coverage and policy communication were also likely elevated, and public attention to the pandemic was probably higher as well.

At the same time, academics may also serve as a kind of early warning indicator. Increased scientific communication could reflect expert concern that is not yet visible in formal policy measures or in mainstream news coverage but that reaches the public through other channels in the days that follow. On Twitter, researchers commented on emerging evidence, often ahead of official policy adoption. As such, the association between scientific messaging and subsequent mobility reflects the fact that academic attention is an early signal in a longer sequence that also includes media coverage, policy deliberation, and eventually behavioral change. This does not diminish the value of monitoring what academics say during health crises. If anything, this just suggests that their public communication contains information about near-term societal responses that is worth taking seriously.

The heterogeneity across cultural contexts is consistent with theoretical expectations. Populations in countries with higher power distance tend to accept hierarchical authority more readily, and those in countries with higher uncertainty avoidance are generally more cautious in the face of unfamiliar risks (Hofstede, 2011). In both settings, expert voices plausibly carry additional weight during a novel health crisis, and this could potentially translate into a stronger link between academic messaging and subsequent behavioral adjustment. These are, to some extent, consistent with existing empirical work. For example, Shan et al. (2024) found that power distance is consistently positively associated with online (Twitter) stances towards COVID-19 preventive measures, including face covering, lockdowns, quarantine, social distancing, and vaccination. Huynh (2020), using country-level OLS estimates from the early weeks of the pandemic, found that uncertainty avoidance appears to be the strongest predictor of social distancing among the three Hofstede dimensions tested (PDI, IDV, and UAI).

In contrast, the moderating role of long-term orientation runs in the opposite direction. One plausible interpretation is that populations in more long-term-oriented settings may already organize daily routines in ways that are more compatible with precautionary adjustment, leaving less room for additional short-term change associated with fluctuations in scientific messaging. In a similar way, some higher-LTO societies may have entered the pandemic with stronger pre-existing public health norms or everyday practices that already aligned behavior with health guidance, thereby reducing the observable association between scientific messaging and subsequent mobility shifts. This is, to some extent, consistent with

the findings of Wang (2021), who found that long-term orientation is associated with a decrease in social distancing during the first wave of the pandemic. Ma et al. (2022) also found that residents in high-LTO countries showed lower willingness to quarantine during the early months of the pandemic in a cross-national survey of 17 countries. They linked this pattern to evidence that long-term-oriented people are less present-biased (Wismans et al., 2021), and that present-biased individuals tend to comply more readily with immediate public health measures (because the salience of the present threat overrides their longer-term preferences).

Individualism shows a similar pattern to long-term orientation, but only for grocery and pharmacy and transit station mobility. The negative association between scientific messaging and population mobility is stronger in more collectivist countries and attenuates in more individualist ones. The theoretical expectation is that collectivist societies may be more disposed to align personal behavior with collective public health goals, including social distancing protocols, whereas individualist societies may value personal autonomy and freedom of movement more strongly (Hofstede, 2011; Chen et al., 2021). For example, Chen et al. (2021) found that individualism moderated the relationship between stay-at-home policies and residential mobility across countries and within the United States and China.

It also appears that workplace mobility, which shows the most consistent direct association with scientific messaging, is moderated by a narrower set of cultural dimensions than other mobility categories. The most plausible explanation is related to the difference between mobility categories in which individuals have behavioral discretion and those in which movement is primarily determined by institutional structures. Workplace attendance falls mainly on the structural side. Whether a person goes to work on a given day is affected more by employer policies, contractual obligations, and formal government directives (office closures, remote work mandates) than by personal interpretation of incoming information. When the scientific messaging share increases and the policy environment tightens, employers across countries are likely to respond in similar ways; that is, they close offices and send people home. That said, the significant PDI and LTO interactions for workplace mobility suggest that even institutional responses may not be entirely uniform across cultural settings.

The categories where cultural moderation is more prominent, such as grocery and pharmacy, retail and recreation, and transit stations, involve a higher degree of individual choice. Whether to postpone a grocery trip or take public transit instead of walking are decisions shaped by personal judgment about risk, appropriateness, and social expectations. These are arguably the spaces where cultural norms may have room to operate. In a society that defers more readily to authority, an individual may follow expert guidance more intently when deciding whether to visit a grocery store. In a society that is more cautious of unfamiliar risks, an individual may be quicker to avoid parks or public spaces when the informational environment signals uncertainty. Workplace mobility may also be less broadly moderated because it involves what is effectively a binary institutional decision (i.e., offices are either open or closed, work is either remote or in-person) rather than the more graded adjustments involved in discretionary travel. How much to reduce grocery trips or how often to visit a park are continuous choices where each person’s threshold for behavioral change is influenced by their normative environment.

Our findings contribute to existing literature through the following. First, they complement existing research on traditional media and mobility during the pandemic by extending

the analysis to direct, unmediated communication by academics on social media. While news coverage remains an important channel, academics on Twitter appear to participate in a parallel flow of information that also tracks with behavioral outcomes. Second, the cross-country design builds on studies that have largely been limited to single national contexts, and the cultural moderation results contribute to a growing literature linking institutional and cultural factors to pandemic response and health outcomes. Third, the findings contribute to ongoing discussions about the role of academics and scientific experts as public communicators in the digital age.

For population health, the practical implication is modest but worth stating. Academic communication is not a substitute for formal public health messaging, and its reach is uneven across disciplines, languages, and platforms. At the same time, our findings suggest that it may function as a useful indicator of early societal engagement with emerging health threats, and possibly as a complementary channel through which science can reach attentive publics during periods of rapid change. How this channel will operate in future crises, and in a post-Twitter social media landscape (Shiffman and Wester, 2025), is an open question. What is clearer is that the line between scientific communication and public behavior is no longer mediated only through institutions and traditional media, and that empirical work on health crises would benefit from treating digital academic discourse as a meaningful part of the communication environment.

In addition, policymakers may find it useful to consider cultural factors in communication and public health strategies to encourage protective behavioral responses (Zidkova et al., 2025; Gokmen et al., 2021). For example, in low-PDI countries, messages may emphasize that measures are part of a shared response to a collective problem rather than an imposition from authority (Gokmen et al., 2021). We caution, however, that these implications are based on the observed moderation of cross-country associations and not from direct evidence that communication strategies tailored to specific cultural profiles produce stronger behavioral responses. Whether message framing adapted to cultural dimensions is associated with the intended behavioral outcomes is still an open question for future research.

7 Limitations

We acknowledge several limitations. First, our analyses identify temporal associations conditional on a substantial set of controls, not causal effects. We do not claim that scientific messaging causes changes in population mobility, and we refer to our estimates throughout as associations net of the epidemiological situation, policy stringency, academic tweeting intensity, country-specific characteristics, and common weekly shocks. Within these limits, our analytical design addresses several common alternative explanations. Country fixed effects absorb time-invariant differences between countries, such as institutional arrangements. Week fixed effects absorb global shocks that moved many countries simultaneously, such as major WHO announcements. We acknowledge, however, that some unobserved factors remain outside our controls. In particular, both scientific messaging and population mobility may respond to the same underlying epidemiological dynamics. When a new wave of infections emerges, academics may shift more of their communication toward COVID-related science, while populations may simultaneously reduce discretionary movement. Our epidemiological controls and fixed effects partially address this concern, but they cannot

capture every dimension of perceived risk, particularly given reporting delays and the gap between official case counts and real-time public perception. That said, the fact that the observed associations vary across mobility categories, lag lengths, and cultural dimensions in theoretically interpretable ways suggests that they are unlikely to be explained entirely by a single common confounding process.

Second, our main independent variable captures the supply side of academic scientific communication. The scientific messaging share measures how prominently COVID-related scientific content appeared within academics’ Twitter activity, but it does not directly measure audience exposure or persuasion. Some academic tweets may have reached broader publics directly or indirectly through journalists, policymakers, and other intermediaries, while others may have circulated within academic networks. We therefore interpret the scientific messaging share as an indicator of the public-facing academic information environment rather than as a direct measure of population exposure to scientific messages. Future research could extend this approach by incorporating engagement, diffusion, or exposure measures to examine how academic communication reaches different audiences.

Third, Twitter data may not be fully representative of the broader scientific community, as the platform tends to be disproportionately used by academics from certain disciplines and by English-speaking researchers. The findings derived from Twitter interactions may therefore not entirely generalize to the entire scientific community as a whole. Our restriction to English-language tweets further limits coverage in non-English-speaking countries, where academic COVID communication in national languages may be both more voluminous and more directly relevant to domestic audiences than the English-language subset we observe. The non-English-speaking robustness check (Section 5.4) provides partial reassurance that these associations are not confined to English-dominant countries. Nevertheless, compared to prior studies (Ke et al., 2017; Mongeon et al., 2023), our approach identified more academics, and Twitter remained, at the time of the study, one of the main channels through which academics engaged in rapid and direct science communication. This makes it a valuable, if partial, window into real-time academic discourse.

Fourth, there are also limitations that must be noted related to our chosen datasets. For example, sampling biases may be associated with the Google Mobility data, which captures movements from Android users who have location history enabled. Despite this, multiple prior studies have successfully used this dataset to analyze aggregate-level societal trends and behaviors (Chan et al., 2023; Okamoto, 2022; Li et al., 2022; Rahman et al., 2021; Devaraj and Patel, 2021).

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